

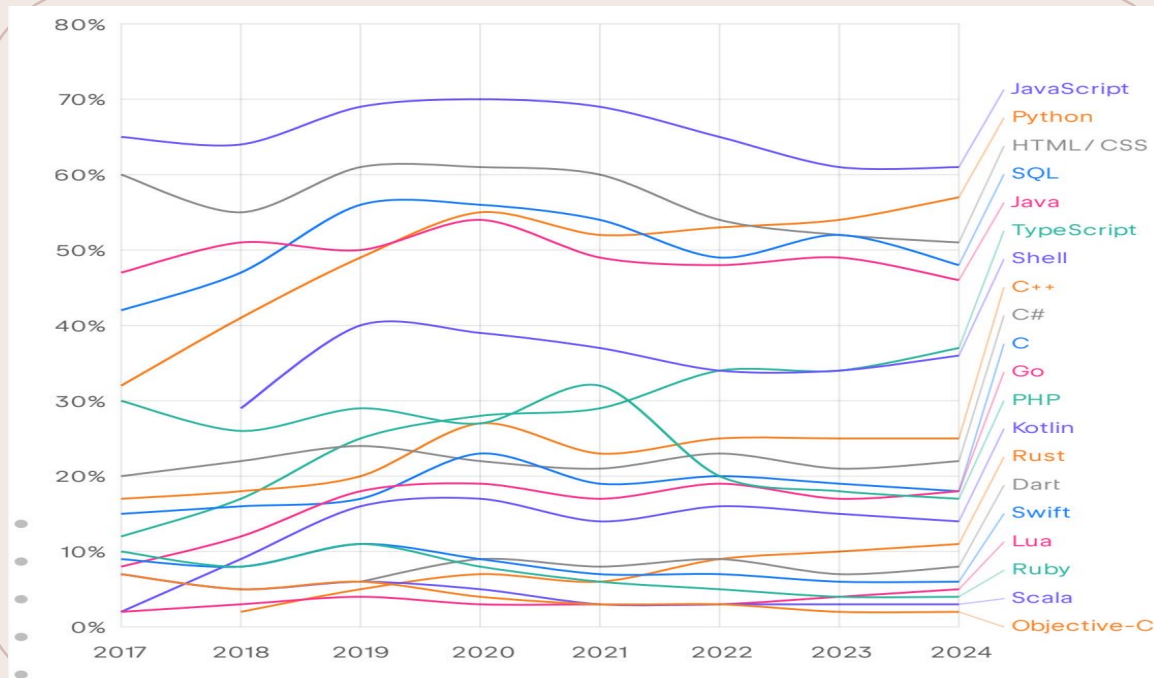
Data Analysis Report

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic

JetBrains' annual survey of 23,262 developers worldwide
2017–2024 language usage trends • satisfaction • adoption intent • domain specialization

Python Closes the Gap: 8 Years of Language Usage Shifts

Usage reshuffles: JavaScript declines from its 2020 peak; Python rises steadily; TypeScript triples; PHP drops sharply.



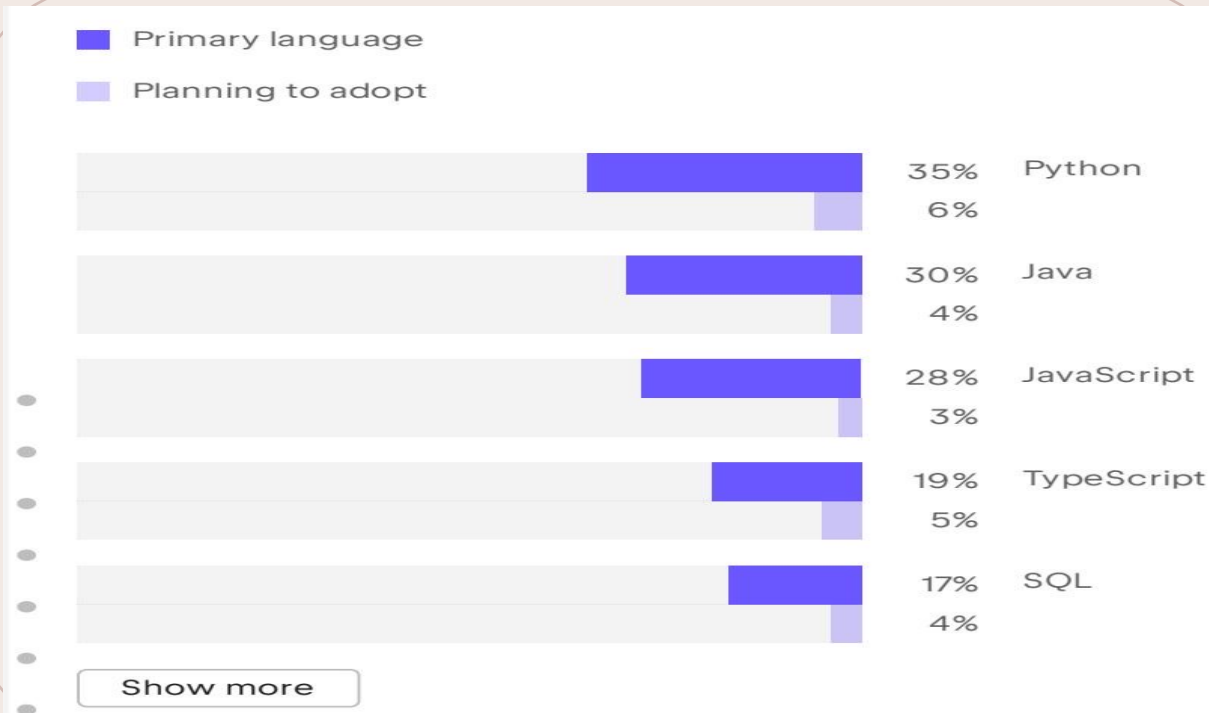
Usage (% of devs)

Language	2017	2020	2022	2024
JavaScript	65	70	65	61
Python	32	55	53	57
HTML/CSS	60	61	54	51
SQL	42	56	49	48
Java	47	54	48	46
TypeScript	12	28	34	37
Go	8	19	19	18
PHP	30	27	20	17

Python Is Now the #1 Primary Language at

35%

Primary language share is concentrated, while adoption intent stays below 6% across languages — a signal of ecosystem lock-in.



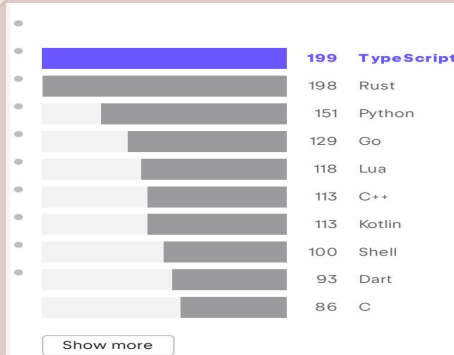
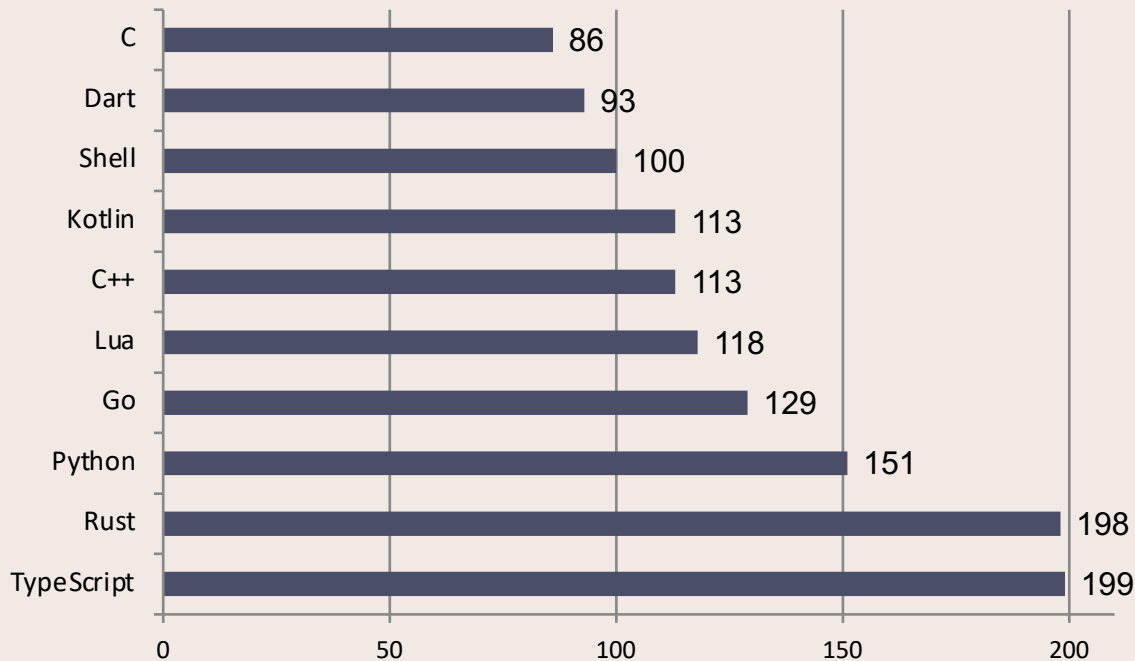
Key figures

Language	Primary	Plan to adopt
Python	35%	6%
Java	30%	4%
JavaScript	28%	—
TypeScript	19%	5%
SQL	17%	4%

Note: Adoption intent is modest ($\leq 6\%$), suggesting developers rarely plan to switch their primary stack.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Top satisfaction languages skew toward safety and modern tooling (scores from JetBrains survey).

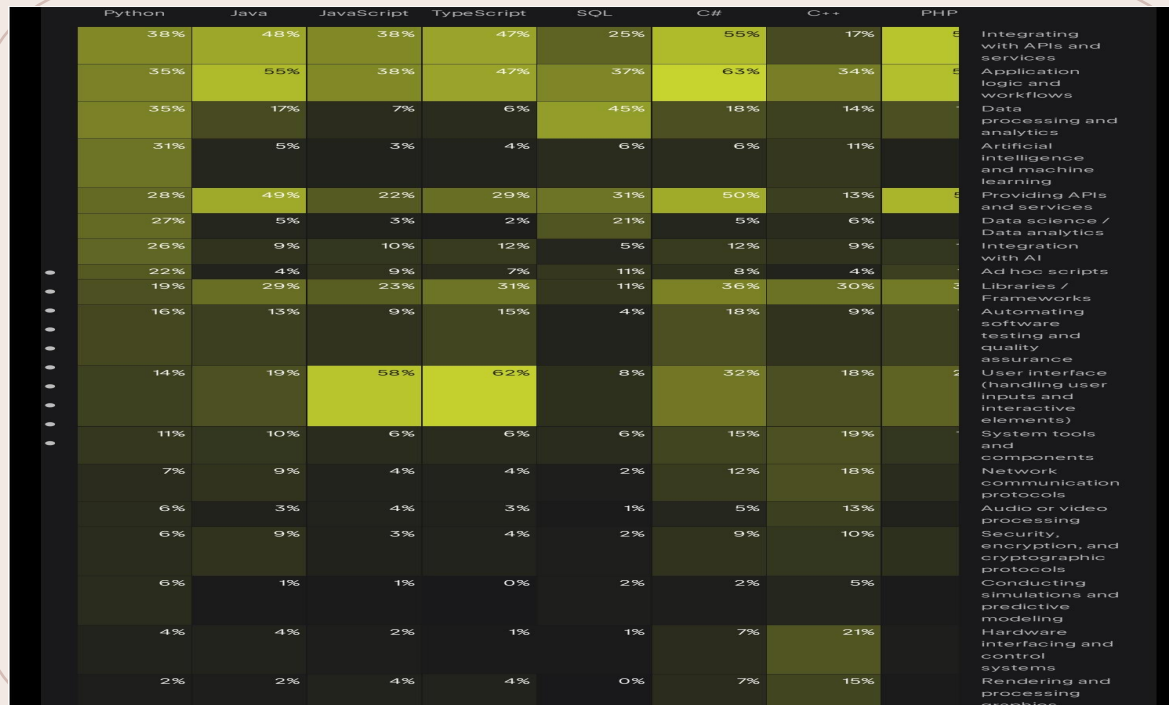


Interpretation

- TypeScript and Rust lead (199/198)
- Python remains strong (151)
- Go follows (129)
- Long tail spans 118→86

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language choice maps to application domains — leaders should plan hiring and platform investments by domain ownership.



Domain leaders

Domain	Top language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating w/ APIs & Services	Python	38%

What This Means for Your Technology Strategy in 2025

Recommendations map directly to measured shifts in usage, satisfaction, and domain leadership.

Strategic actions

- 1) Invest in Python for AI/ML teams — it leads AI/ML (31%) and data science (28%).
- 2) Treat TypeScript as the frontend + full-stack default — it dominates UI (62%) and leads satisfaction (199).
- 3) Keep Rust and Go on the infra roadmap — Rust satisfaction (198) and sustained Go usage (18%) support modern systems needs.
- 4) Expect shrinking pools in declining ecosystems — PHP drops from 30% (2017) to 17% (2024); Ruby from 10% to 4%.
- 5) Use developer satisfaction as a retention input — top languages emphasize safety + tooling.

Headline shift

Python is #1 primary language (35%) while JavaScript remains #1 overall usage (61%).

Developer preference

Satisfaction leaders: TypeScript (199) and Rust (198).

Domain reality

AI/ML → Python (31%)
UI → TypeScript (62%)
APIs → Java (49%)
Systems → C++ (27%)